Section 5.1 : #1.(a)~(k), #3.(b),(d), #4.(e),(g),(j), #11, #14, #15.(a), #16.(a), #17

*Click the problem number above.

1. Label the following statements as true or false.

(a) Every linear operator on an n-dimensional vector space has n distinct eigenvalues.

Solution: False

Let us give a counterexample. Consider the identity matrix I (or the identity map Id). It has a unique eigenvalue 1.

(b) If a real matrix has one eigenvector, then it has an infinite number of eigenvectors.

Solution: True

The eigenvector v of a linear operator T corresponding to the eigenvalue λ is the solution of the homogenous system of linear equations $(T-\lambda I)x=0$ by Theorem 5.4. Recall that the homogeneous system has either a unique solution 0 or infinitely many solutions. Since eigenvectors are nonzero by definition, there is a nonzero solution $v\neq 0$.

(c) There exists a square matrix with no eigenvectors.

Solution: True

Let us give an example. Consider a matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_{2\times 2}(\mathbb{R})$. There is no real eigenvalue and so no eigenvector.

(d) Eigenvalues must be nonzero scalars.

Solution: False

Let us give a counterexample. Consider the zero matrix 0 (or the zero map). It has a unique eigenvalue 0.

(e) Any two eigenvectors are linearly independent.

Solution: False

Let us give a counterexample. Consider the identity matrix I (or the identity map Id). It has infinitely many linearly dependent eigenvectors $\{(t, \ldots, t) \mid t \in F\}$ corresponding to the eigenvalue 1.

(f) The sum of two eigenvalues of a linear operator T is also an eigenvalue of T.

Solution: False

Let us give a counterexample. Consider a matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \in M_{2\times 2}(\mathbb{R})$. It has two eigenvalues $1, -1$, but $0 = 1 - 1$ is not an eigenvalue.

(g) Linear operators on infinite-dimensional vector spaces never have eigenvalues.

Solution: False

Let us give a counterexample. Consider the identity map Id which has an eigenvalue 1.

(h) An $n \times n$ matrix A with entries from a field F is similar to a diagonal matrix if and only if there is a basis for F^n consisting of eigenvectors of A.

Solution: True

Let A be similar to a diagonal matrix D. That is, $D = Q^{-1}AQ$ for some invertible matrix Q. Then the columns of Q will be eigenvectors of A forming a basis for F^n. Conversely, n linearly independent eigenvectors of A form the invertible matrix Q such that $D = Q^{-1}AQ$ for some diagonal matrix D.

(i) Similar matrices always have the same eigenvalues.

Solution: True

Let A, B be similar matrices such that $A = Q^{-1}BQ$ for some invertible matrix Q.

It follows from $\det(B - \lambda I) = \det\left(Q^{-1}(B - \lambda I)Q\right) = \det\left(Q^{-1}BQ - \lambda I\right) = \det(A - \lambda I)$.

(j) Similar matrices always have the same eigenvectors.

Solution: False

Let us give a counterexample. Consider two similar matrices $A=\begin{pmatrix}4&1\\1&1\end{pmatrix}$ and $B=\begin{pmatrix}1&1\\1&4\end{pmatrix}$ such that $B=\begin{pmatrix}0&1\\1&0\end{pmatrix}A\begin{pmatrix}0&1\\1&0\end{pmatrix}$.

(k) The sum of two eigenvectors of an operator T is always an eigenvector of T.

Solution: False

Let us give a counterexample. Consider the matrix $A=\begin{pmatrix}1&0\\0&-1\end{pmatrix}$ and its two eigenvectors $\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix}$. But their sum is not the eigenvector for A.

3. For each of the following matrices $A \in \mathsf{M}_{n\times n}(F)$,

(i) Determine all the eigenvalues of A.

(ii) For each eigenvalue λ of A, find the set of eigenvectors corresponding to λ.

(iii) If possible, find a basis for F^n consisting of eigenvectors of A.

(iv) If successful in finding such a basis, determine an invertible matrix Q and a diagonal matrix D such that $Q^{-1}AQ = D$.

*The answer for this problem is not unique.

(b) $A = \begin{pmatrix}0&-2&-3\\-1&1&-1\\2&2&5\end{pmatrix}$ for $F = R$

Answers:

eigenvalues: $\lambda = 1, 2, 3$

eigenvectors: $\begin{pmatrix}-1\\-1\\1\end{pmatrix}$ $(\lambda=1)$, $\begin{pmatrix}-1\\1\\0\end{pmatrix}$ $(\lambda=2)$, $\begin{pmatrix}-1\\0\\1\end{pmatrix}$ $(\lambda=3)$

basis: $\begin{pmatrix}-1\\-1\\1\end{pmatrix}$, $\begin{pmatrix}-1\\1\\0\end{pmatrix}$, $\begin{pmatrix}-1\\0\\1\end{pmatrix}$

invertible matrix & diagonal matrix: $Q=\begin{pmatrix}-1&-1&-1\\-1&1&0\\1&0&1\end{pmatrix}$, $D=\begin{pmatrix}1&0&0\\0&2&0\\0&0&3\end{pmatrix}$

<u>Solution:</u>

Consider the characteristic polynomial $\det(A-\lambda I) = -(\lambda-1)(\lambda-2)(\lambda-3)=0$. This shows that the eigenvalues are $1,2,3$.

For $\lambda=1$, consider the equation $(A-\lambda I)x = \begin{pmatrix}-1&-2&-3\\-1&0&-1\\2&2&4\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} = 0$. Then the solution is $t\begin{pmatrix}-1\\-1\\1\end{pmatrix}$ for all $t\in\mathbb{R}$. Similary, we can find all the other eigenvectors $\begin{pmatrix}-1\\1\\0\end{pmatrix}$, $\begin{pmatrix}-1\\0\\1\end{pmatrix}$ corresponding to $\lambda=2,3$ respectively.

Since three eigenvectors are linearly independent, they form a basis and induce an invertible matrix $Q=\begin{pmatrix}-1&-1&-1\\-1&1&0\\1&0&1\end{pmatrix}$ such that $D= Q^{-1}AQ=\begin{pmatrix}1&0&0\\0&2&0\\0&0&3\end{pmatrix}$.

(d) $A = \begin{pmatrix}2&0&-1\\4&1&-4\\2&0&-1\end{pmatrix}$ for $F = R$

<u>Answers:</u>

eigenvalues: $0,1$

eigenvectors: $\begin{pmatrix}1\\4\\2\end{pmatrix}$ $(\lambda=0)$, $\begin{pmatrix}0\\1\\0\end{pmatrix}$ $(\lambda=1)$, $\begin{pmatrix}1\\0\\1\end{pmatrix}$ $(\lambda=1)$

basis: $\begin{pmatrix}1\\4\\2\end{pmatrix}$, $\begin{pmatrix}0\\1\\0\end{pmatrix}$, $\begin{pmatrix}1\\0\\1\end{pmatrix}$

invertible matrix & diagonal matrix: $Q=\begin{pmatrix}1&0&1\\4&1&0\\2&0&1\end{pmatrix}$, $D=\begin{pmatrix}0&0&0\\0&1&0\\0&0&1\end{pmatrix}$

Solution:
We omit this since the answers can be obtained as in Exercise 3(b) above.

4. For each linear operator T on V, find the eigenvalues of T and an ordered basis β for V such that $[\mathsf{T}]_\beta$ is a diagonal matrix.

*The answer for this problem is not unique.

(e) $\mathsf{V} = \mathsf{P}_2(R)$ and $\mathsf{T}(f(x)) = xf'(x) + f(2)x + f(3)$

Solution:
Note that

$$\begin{array}{cccc} T: & P_2(R) & \to & P_2(R) \\ & (ax^2+bx+c) & \mapsto & 2ax^2+(4a+3b+c)x+9a+3b+c \end{array}.$$

If we choose a basis $\gamma = \{x^2, x, 1\}$ for T, then $A = [T]_\gamma = \begin{pmatrix} 2 & 0 & 0 \\ 4 & 3 & 1 \\ 9 & 3 & 1 \end{pmatrix}$. So the characteristic polynomial is $\det(A-\lambda I) = -\lambda(\lambda-2)(\lambda-4) = 0$, and the eigenvalues are $0, 2, 4$.

To find the eigenvectors, let us solve the equation $(A-\lambda I)x = 0$ for each $0, 2, 4$. Then we get $\begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} -4 \\ 13 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ as corresponding eigenvectors respectively. Hence the desired basis is

$\beta = \{-x+3, -4x^2+13x+3, x+1\}$ and $[T]_\beta = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ by Theorem 5.1.

(g) $\mathsf{V} = \mathsf{P}_3(R)$ and $\mathsf{T}(f(x)) = xf'(x) + f''(x) - f(2)$

Solution:
Note that

$$\begin{array}{cccc} T: & P_3(R) & \to & P_3(R) \\ & (ax^3+bx^2+cx+d) & \mapsto & 3ax^3+2bx^2+(6a+c)x-(8a+2b+2c+d) \end{array}.$$

If we choose a basis $\gamma = \{x^3, x^2, x, 1\}$ for T, then $A = [T]_\gamma = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -8 & -2 & -2 & -1 \end{pmatrix}$. So the characteristic polynomial is $\det(A-\lambda I) = (\lambda+1)(\lambda-1)(\lambda-2)(\lambda-3)$, and the eigenvalues are $-1, 1, 2, 3$.

To find the eigenvectors, let us solve the equation $(A-\lambda I)x=0$ for each $-1,1,2,3$. Then we get $\begin{pmatrix}0\\0\\0\\1\end{pmatrix}, \begin{pmatrix}0\\0\\-1\\1\end{pmatrix}, \begin{pmatrix}0\\-3\\0\\2\end{pmatrix}, \begin{pmatrix}-2\\0\\-6\\7\end{pmatrix}$ as corresponding eigenvectors respectively. Hence the desired basis is $\beta=\{1, -x+1, -3x^2+2, -2x^3-6x+7\}$ and $[T]_\beta=\begin{pmatrix}-1&0&0&0\\0&1&0&0\\0&0&2&0\\0&0&0&3\end{pmatrix}$ by Theorem 5.1.

(j) $\mathsf{V}=\mathsf{M}_{2\times 2}(R)$ and $\mathsf{T}(A)=A^t+2\cdot \mathrm{tr}(A)\cdot I_2$

Solution:
Note that

$$\begin{array}{cccc} T: & M_{2\times 2}(R) & \to & M_{2\times 2}(R) \\ & \begin{pmatrix}a&b\\c&d\end{pmatrix} & \mapsto & \begin{pmatrix}3a+2d & c\\ b & 2a+3d\end{pmatrix}\end{array}.$$

If we choose a basis $\gamma=\left\{\begin{pmatrix}1&0\\0&0\end{pmatrix}, \begin{pmatrix}0&1\\0&0\end{pmatrix}, \begin{pmatrix}0&0\\1&0\end{pmatrix}, \begin{pmatrix}0&0\\0&1\end{pmatrix}\right\}$ for T, then $A=[T]_\gamma=\begin{pmatrix}3&0&0&2\\0&0&1&0\\0&1&0&0\\2&0&0&3\end{pmatrix}$. So the characteristic polynomial is $\det(A-\lambda I)=(\lambda+1)(\lambda-1)^2(\lambda-5)$, and the eigenvalues are $-1,1,5$.

To find the eigenvectors, let us solve the equation $(A-\lambda I)x=0$ for each $-1,1,5$. Then we get $\begin{pmatrix}0\\-1\\1\\0\end{pmatrix}, \begin{pmatrix}0\\1\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\0\\1\end{pmatrix}, \begin{pmatrix}1\\0\\0\\1\end{pmatrix}$ as corresponding eigenvectors to $-1,1,1,5$ respectively. Hence the desired basis is $\beta=\left\{\begin{pmatrix}0&-1\\1&0\end{pmatrix}, \begin{pmatrix}0&1\\1&0\end{pmatrix}, \begin{pmatrix}-1&0\\0&1\end{pmatrix}, \begin{pmatrix}1&0\\0&1\end{pmatrix}\right\}$ and $[T]_\beta=\begin{pmatrix}-1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&5\end{pmatrix}$ by Theorem 5.1.

11. A **scalar matrix** is a square matrix of the form λI for some scalar λ; that is, a scalar matrix is a diagonal matrix in which all the diagonal entries are equal.

(a) Prove that if a square matrix A is similar to a scalar matrix λI, then $A = \lambda I$.

Solution:

If A is similar to λI, then $A = Q^{-1}(\lambda I)Q = \lambda I$.

(b) Show that a diagonalizable matrix having only one eigenvalue is a scalar matrix.

Solution:

If any diagonalizable matrix has one eigenvalue, then it is similar to a scalar matrix. Hence by (a), it is just a scalar matrix.

(c) Prove that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not diagonalizable.

Solution:

Suppose it is diagonalizable. Since it has one eigenvalue 1, it should be a scalar matrix by (b). Contradiction.

14.† For any square matrix A, prove that A and A^t have the same characteristic polynomial (and hence the same eigenvalues).

Solution:

Note that $\det(A - \lambda I) = \det(A - \lambda I)^t = \det(A^t - \lambda I)$.

15.† **(a)** Let T be a linear operator on a vector space V, and let x be an eigenvector of T corresponding to the eigenvalue λ. For any positive integer m, prove that x is an eigenvector of T^m corresponding to the eigenvalue λ^m.

Solution:

Let us use the induction on $m \in N$.

If $m = 1$, then it is clear that $T(x) = \lambda x$ since x is the eigenvector corresponding to λ.

Assume that it is true for all $m \le i$. Then it is also true for $m = i+1$ since $T^{i+1}(x) = T\left(T^i(x)\right) = T\left(\lambda^i x\right) = \lambda^{i+1}x$.

16. (a) Prove that similar matrices have the same trace. *Hint:* Use Exercise 13 of Section 2.3.

Solution:

Let A, B be similar matrices such that $A = Q^{-1}BQ$ for some invertible matrix Q. Then $tr(A) = tr\left(Q^{-1}BQ\right) = tr\left(BQQ^{-1}\right) = tr(B)$ by Exercise 13 of Section 2.3.

17. Let T be the linear operator on $\mathsf{M}_{n\times n}(R)$ defined by $\mathsf{T}(A) = A^t$.

(a) Show that ± 1 are the only eigenvalues of T.

Solution:

Let A be the eigenvector corresponding to the eigenvalue λ. Since $T^2(A) = A = \lambda^2 A$ and A is nonzero, we have $\lambda^2 = 1$ so $\lambda = \pm 1$.

(b) Describe the eigenvectors corresponding to each eigenvalue of T.

Solution:

By (a), there are two cases: $A = A^t\,(\lambda = 1)$ or $A = -A^t\,(\lambda = -1)$. This shows that the eigenvectors corresponding to 1 are symmetric matrices, and the eigenvectors corresponding to -1 is skew-symmetric matrices.

(c) Find an ordered basis β for $\mathsf{M}_{2\times 2}(R)$ such that $[\mathsf{T}]_\beta$ is a diagonal matrix.

Solution:

Let us consider eigenvectors of $1, -1$ by (b).

If $\lambda = 1$, we need to find a basis for the subspace consisting of symmetric matrices.

Hence we choose three eigenvectors $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

And if $\lambda=-1$, we need to find a basis for the subspace consisting of skew-symmetric matrices.

Hence we choose one eigenvector $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

These 4 vectors we chose will form a basis β such that $[T]_\beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$.

(d) Find an ordered basis β for $\mathsf{M}_{n\times n}(R)$ such that $[\mathsf{T}]_\beta$ is a diagonal matrix for $n > 2$.

Solution:

In general, it suffices to find a basis for symmetric matrices and a basis for skew-symmetric matrices. For convenience, denote by E_{ij} the matrix in which the only nonzero entry is a 1 in the ith row and jth column. Then the following eigenvectors will form our desired basis.

(1) ($\lambda=1$) E_{ii} for all $1 \le i \le n$

(2) ($\lambda=1$) $E_{ij}+E_{ji}$ for all $1 \le i \le n$ and $i<j$

(3) ($\lambda=-1$) $E_{ij}-E_{ji}$ for all $1 \le i \le n$ and $i<j$